

Sterilization & Disinfection



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❑ **Sterilization** : is the freeing of an article of all micro-organisms (vegetative form & spores) .

❑ **Disinfection** : is the destruction of pathogenic micro-organisms, but not necessarily spores.

- Bacteria differ in their response to physical and chemical conditions.
- The sensitivity of any organism to a killing agent is known as *the organism innate susceptibility.*
- Several other factors, besides the innate susceptibility of the organism, influence the action of the killing agent.

- 1- The number of the organisms present .**
- 2- The time necessary for action.**
- 3- The concentration of the killing agent.**
- 4- The temperature.**
- 5- The optimum acidity or alkalinity for activity of the killing agent .**

6- Organic matters e.g. Blood, pus and mucus act as a protective coat for the bacteria and limit the action of killing agents.

7- Direct contact between all surfaces of the object to be sterilized and the killing agent. Air bubbles in tubing will protect bacteria from steam or chemicals.

8- Presence of antagonistic substances neutralizing the activity of the killing agent.

**Number of
the organisms**

**The concentration
of killing agent**

The time for action

The temperature

Factors affecting the action of the killing agent

Organic matters

Acidity , alkalinity

Direct contact

**Antagonistic
substances**

Cleaning

As the number of organisms present affects the sterilizing time and efficiency :

- It is very important to reduce the number before attempting to apply any method of sterilization.
- It is also essential to remove organic material which will protect bacteria against the action of physical or chemical agents by preventing direct contact.

➤ Careful washing of all equipments or running hot water (below 60°C) with soap or detergent is highly recommended before the process of sterilization.

➤ The agents used for sterilization and disinfection are divided into physical and chemical agents.

Microbial Control Methods

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graph TD; A[Microbial Control Methods] --> B[Physical Agents]; A --> C[Chemical Agents];
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Physical Agents

Chemical Agents

Methods of Sterilization:

- Methods of sterilization are:

1- Physical Methods:

- a. Radiation method .
- b. Filtration method .
- c. Thermal (Heat) methods:

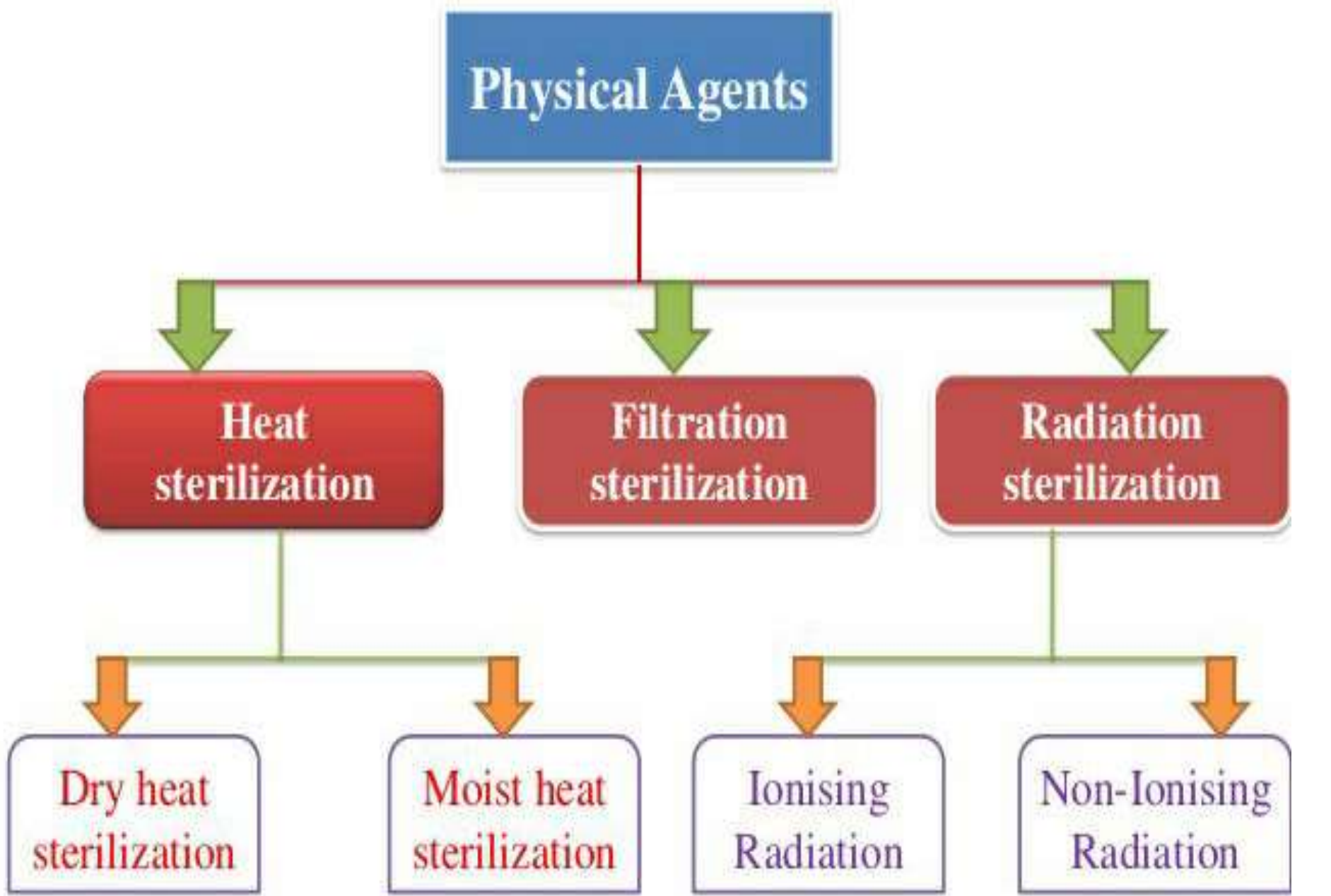
I- Dry Heat.

II- Moist Heat.

2- Chemical Method:

In this method we use various chemicals.

Physical Agents



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graph TD; A[Physical Agents] --> B[Heat sterilization]; A --> C[Filtration sterilization]; A --> D[Radiation sterilization]; B --> E[Dry heat sterilization]; B --> F[Moist heat sterilization]; D --> G[Ionising Radiation]; D --> H[Non-Ionising Radiation];
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The diagram is a hierarchical flowchart. At the top is a blue box labeled 'Physical Agents'. A red line descends from this box and splits into three green arrows pointing to three red boxes: 'Heat sterilization', 'Filtration sterilization', and 'Radiation sterilization'. From 'Heat sterilization', a green line descends and splits into two orange arrows pointing to 'Dry heat sterilization' and 'Moist heat sterilization'. From 'Radiation sterilization', a green line descends and splits into two orange arrows pointing to 'Ionising Radiation' and 'Non-Ionising Radiation'. 'Filtration sterilization' has no further sub-categories shown.

**Heat
sterilization**

**Filtration
sterilization**

**Radiation
sterilization**

**Dry heat
sterilization**

**Moist heat
sterilization**

**Ionising
Radiation**

**Non-Ionising
Radiation**

Physical agents

Heat

Heat may be used either as dry or as moist heat (either with or without the presence of Water or water vapor).

So destruction of micro-organisms by heat can be achieved either by:

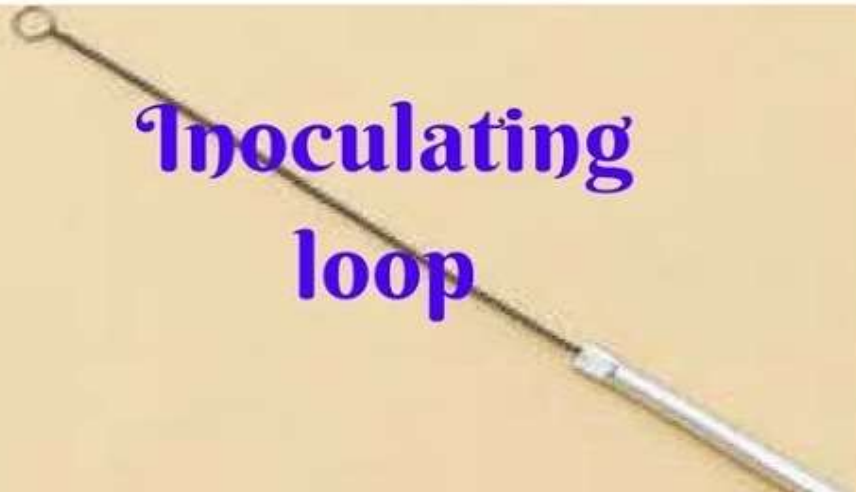
- Rapid coagulation of the cell protein (**moist heat**).
- Slower process of destructive oxidation (**dry heat**).

Differences between dry heat sterilization and moist heat sterilization

Moist Heat sterilization	Dry Heat sterilization
The moist heat sterilization have water and steam.	The dry heat sterilization have no use of water and steam.
Sterilization with coagulation of protein.	Sterilization with oxidation.
This process is under pressure.	This process is on direct flame.
This process takes less time.	This process takes more time.
Moist heat are boiling and autoclave.	Dry heat are flame and incineration.



Dry Heat Sterilization



Inoculating
loop



Oven

Dry heat

Is less efficient than moist heat ; High temperature , long exposure times are required.

a-Direct application of flame .

b-Hot air oven.

c-Infrared sterilizer.

a-Direct application of flame :

- ❑ **Wire loops** used in bacteriology laboratory sterilized by heating to redness in Bunsen burner flame.
- ❑ **Incineration** is the safest method of sterilizing , disposing infected dressings.



Incinerator



b-Hot air oven :

☐ Sterilization of metal instruments , glassware, syringes ,oily preparation as Vaseline , sulfa drugs and talk powder.

☐ The standard temperature is 160 °C.

☐ The time of exposure is 60 min.

c-Infrared sterilizer:

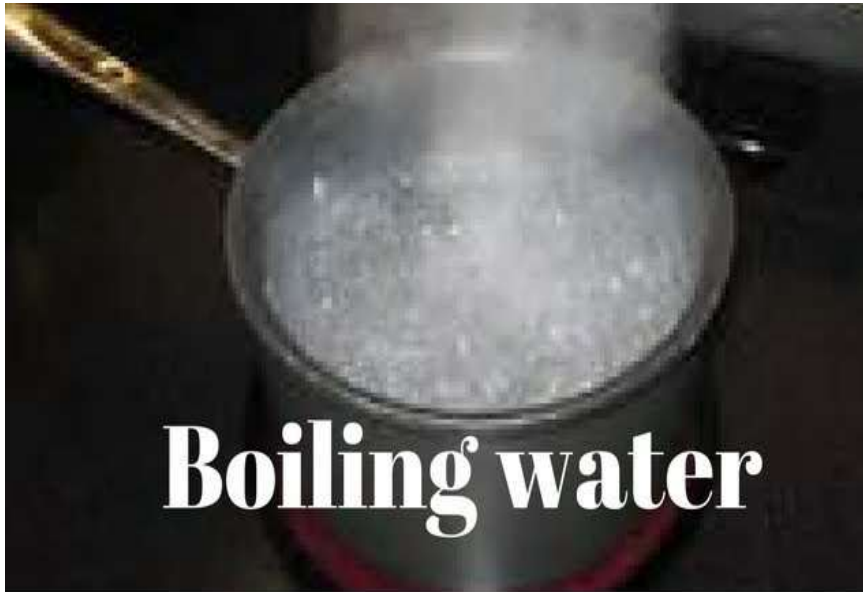
- ❑ Infra-red rays directed onto the objects to be sterilized .
- ❑ Sterilization of metal instruments.
- ❑ Temperature is 180 °C.
- ❑ The time of exposure is 16 min.

Hot air oven



Infrared sterilizer





Boiling water



Autoclave



Moist

HEAT

Sterilization

Moist heat

Relatively low temperature will destroy micro-organism in presence of moisture.

a-Pasteurization .

b-Boiling .

c- Autoclave.

a-Pasteurization :

✓ Makes milk safe for human consumption by destroying pathogenic bacteria as T.B. ,brucella. It is not suitable for spores destruction.

✓ There are two methods :

The holder method : raising the temperature of the milk to **63° C** for **30 min**. then cooling quickly to 10°C or below.

The flash method : raising the temperature to **74°C** for **5 sec**. then cooling quickly to 10°C or below.

b-Boiling :

- ✓ **Total immersion in boiling water at 100°C for 10 min. , is a simple and rapid method to kill most pathogenic bacteria.**
- ✓ **But it does not kill all bacteria, spores even if boiling continued for several hours.**
- ✓ **It cannot therefore said to be a reliable sterilizing method.**
- ✓ **It should be used if alternative sterilizing methods are not available.**

Pasteurization



Boiling



c- Autoclave:

- ✓ This is a device objects heated in the presence of saturated steam at pressure higher than atmospheric.
- ✓ The pressure usually used is **double atmospheric**.
- ✓ The temperature is **121°C**.
- ✓ The exposure time required is **15-20 min**.

*** It is important that the autoclave should contain *steam without air* because :**

- 1. It prevents the steam from entering porous objects as dressings.**
- 2. It reduces effectiveness of steaming as a killing agent.**

There are many types of the autoclave:

- **Simple** metallic container with air-tight lid, pressure gauge, steam tap and safety valve manually operated.
- **Highly complex** automatic machine (high vacuum autoclave).



Types of autoclave

Physical agents

Radiation

- 1- Ultraviolet light.
- 2- Ionizing radiation.

1- Ultraviolet light:

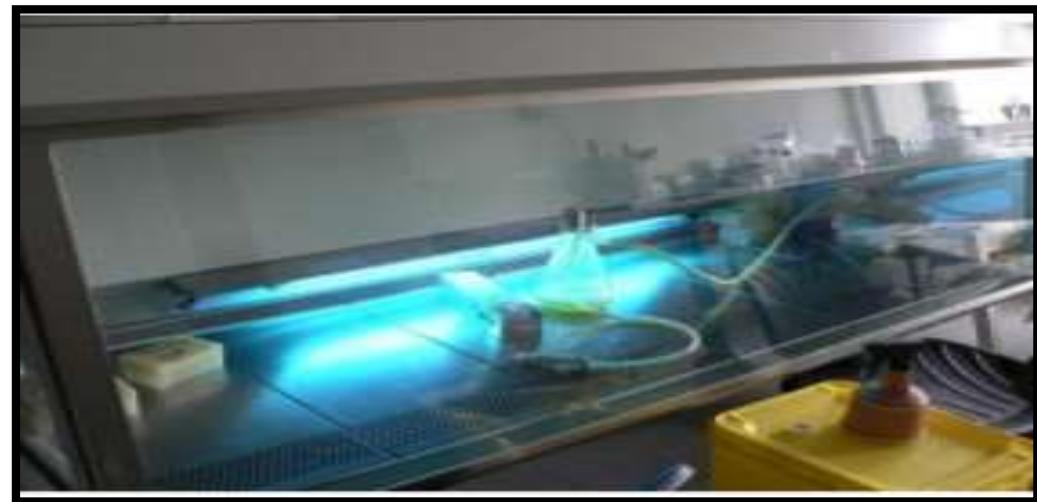
***Prolonged exposure to fresh air and sunlight is perhaps the oldest and most pleasant anti-infection measure.**

***Ultraviolet rays for disinfection of air in rooms is of little practical value and has many limitations.**

*** Ultraviolet rays does not penetrate.**

*** Its lethal effect on bacteria is not instantaneous فوريا i.e. it needs adequate long exposure.**

*** Direct exposure of man to ultraviolet light causes many side effects e.g. conjunctiva damage**



Ultraviolet radiation

2-Ionizing radiation:

- * The sources of ionizing radiations is radioactive isotope cobalt 60.**
- * Its action is interfering with the enzymatic activity leading to cell death.**
- * High penetration power.**
- * Safe sterilization for disposable plastic syringes and catheters.**

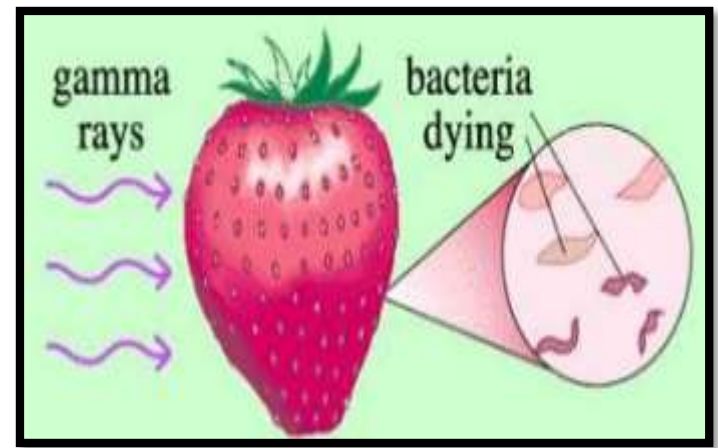
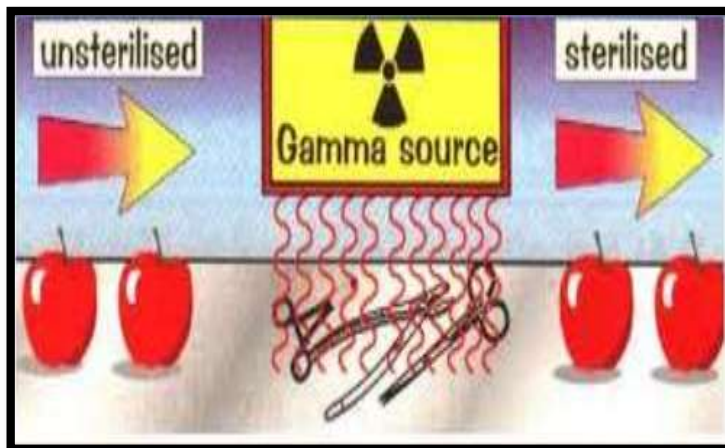
Radioactive Isotope



Gamma Ray



- Most Popular
- Simple, Easily Controlled, Widely Applicable



Physical agents

Filtration

***The principle :** pass the material to be sterilized through special filters which hold back any bacteria present; the filtrate is thus obtained bacteria-free.

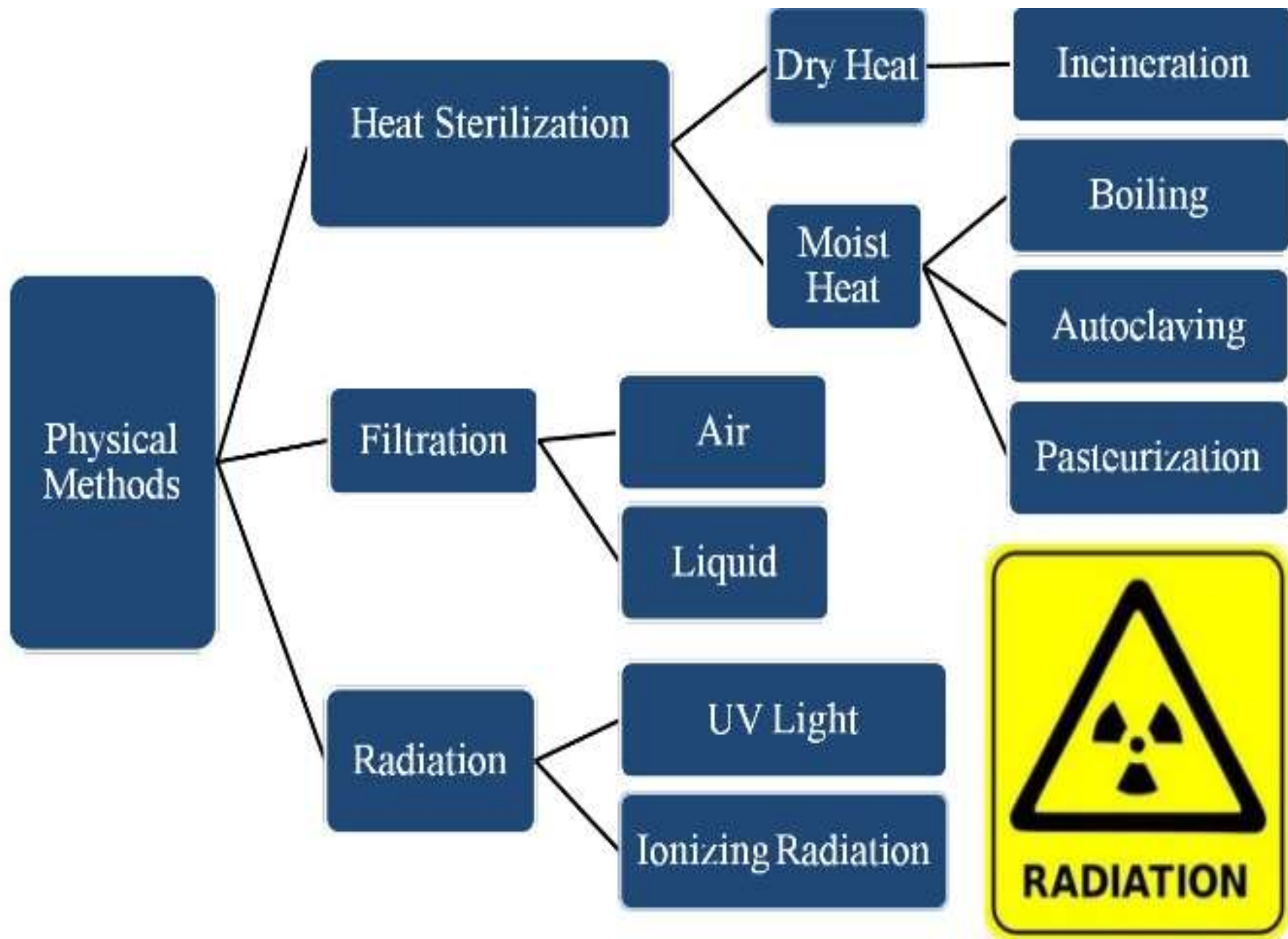
***Before use the filter and the container meet to receive the filtrate are all sterilized in the autoclave.**

- * Filtration may be carried out in:
 - **Seitz filter** using a thick pad of corn-pressed asbestos to retain the bacteria.
 - **Modern filters** which use thin porous sheets of a cellulose ester.

* Filtration is used for the sterilization of fluids that do not stand heating (**thermolabile fluids**)
e.g. serum, plasma, vitamins.

Filtration





Chemical agents

* All chemical preparations available on the market are less reliable as sterilizing agents than heat or radiation.

➤ The term disinfectant usually describes a chemical agent capable of killing micro-organisms, but toxic and suitable only for application on **inanimate objects**.

➤ The term antiseptic is used for weaker and less toxic agents which inhibit and may kill micro-organisms if and can be superficially applied to **living tissues**.

Chemical agents

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graph TD; A([Chemical agents]) --- B([Liquids]); A --- C([Gases]);
```

Liquids

Gases

Liquids

➤ Alcohol

➤ Phenol and Cresol compounds

➤ Halogens

➤ Heavy metals

➤ Oxidizing agents

➤ Quaternary ammonium compounds



❑ **Alcohol:** Ethyl alcohol 70% is an effective antiseptic for the **skin**. Greater concentrations are less effective since water is essential to the transfer of the antiseptic across the bacterial cell wall.

❑ **Phenol and Cresol compounds:**

- * **Phenol** (carbolic acid) is potent disinfectant.
- * **Cresol compound** (Lysol) is an effective, cheap disinfectant for floor and linen.

❑ **Halogens:** Iodine 2.5% in 70% alcohol is the most effective skin germicide but irritant, cause a severe reaction in a sensitive subject and has the disadvantage of staining.

■ **Iodophores:** (Betadine) are combinations of iodine and non-ionic detergents. They have the good properties of iodine but non-staining and less irritant.

■ **Chloramine compounds** such as (chlorinated lime) (domestic bleach) is used for disinfection of swimming pools. (Eusol) is a chlorinated lime and boric acid compound used for wound dressing.

❑ **Heavy metals:** Silver nitrate 1 % used in treatment of burns .

❑ **Oxidizing agents:** Hydrogen peroxide is an antiseptic liberates oxygen when in contact with organic material. The effervescence helps mechanically to loosen slough and pus in wounds.

❑ **Quaternary ammonium compounds:** Cetavlon ,Cetrimide limited effect on Gram - negative bacteria and no effect on T.B.



Liquids



Liquids



Gases

- **Formaldehyde gas.**
- **Ethylene oxide gas.**

Formaldehyde gas:

- ✓ Used for terminal room disinfection and for objects which will not stand high temperature e.g. leather goods and books.
- ✓ The best results obtained with **high concentrations of the gas**; **humidity above 60 %** and a temperature not less than 18°C.
- ✓ The main disadvantage is that it is irritant and has a pungent (لاذع) smell.

Ethylene oxide gas:

Used for sterilization of heat sensitive materials e.g. plastic and rubber articles and cat gut.

Formaldehyde gas

Ethylene oxide gas



Gases



Choose the correct answer:

1) Infrared sterilizer is a method of :

- Heat sterilization.
- Chemical sterilization.
- Sterilization by radiation.

2) A method of dry heating :

- Autoclave.
- Hot air oven
- pasteurization.

3) Plastic syringes are sterilized by :

- Ionizing radiation.
- Autoclave.
- Hot air oven.

4) Talk powder is sterilized by :

- Boiling.
- Hot air oven.
- Autoclave.

5) Seitz Filter is suitable for sterilization of :

- Vitamins.
- Talk powder.
- Milk.

True or false:

1- Sulfa drugs can be sterilized by hot air oven.

2- Autoclave is a type of moist heating.

3- Rubber catheters can be sterilized by ethylene oxide.

4- Ultraviolet rays have high penetration power.

5- Seitz filter can be sterilized by autoclave.

Fill in the space :

- 1- Killing agents may be& agents.**
- 2- Heat methods include heat & heat.**
- 3- can be applied to inanimate objects, while can be applied to living tissues.**
- 4- is the sensitivity of any organism to the killing agent.**
- 5- is the freeing of an article of all microorganisms whether vegetative or sporulated form.**

